



FACULTÉ DES SCIENCES ET DES TECHNOLOGIES(FST)

TD N°9_ – RÉSEAUX 2

COURS: RÉSEAUX 2

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L'objectif de ce TD est de:

- 1- Installer et configurer Snort
- 2- Activer Snort sur l'interface WAN
- 3- Détecter une attaque réseau
- 4- Analyser les alertes générées
- 5- Comprendre le fonctionnement IDS vs IPS

Installation de Snort

The screenshot shows the pfSense web interface. The browser address bar indicates the URL is 192.168.2.1/pkg_mgr.php. The page title is "System / Package Manager / Available Packages". Below the title, there are two tabs: "Installed Packages" and "Available Packages", with the latter being selected. A search bar is present with the text "Snort" entered. Below the search bar, a table lists the available packages. The table has three columns: "Name", "Version", and "Description". The first row shows "snort" with version "4.1.6_28". The description states: "Snort is an open source network intrusion prevention and detection system (IDS/IPS). Combining the benefits of signature, protocol, and anomaly-based inspection." To the right of the description is a green button with a plus sign and the text "Install". Below the table, there is a section for "Package Dependencies:" which lists "snort-2.9.20_8".

pfSense
COMMUNITY EDITION

System / Package Manager / Available Packages

Installed Packages Available Packages

Search

Search term: Snort Both Search Clear

Enter a search string or *nix regular expression to search package names and descriptions.

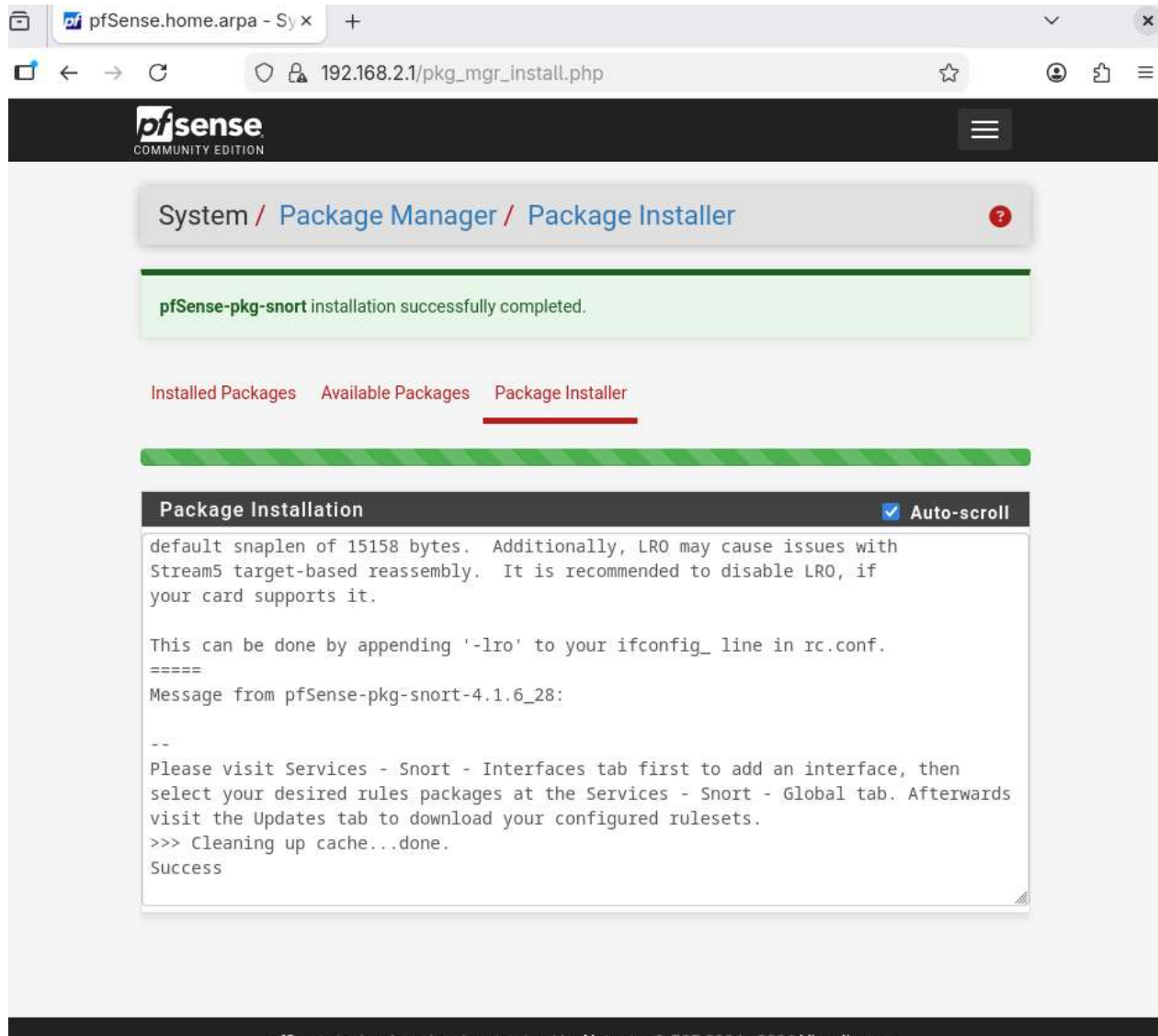
Packages

Name	Version	Description
snort	4.1.6_28	Snort is an open source network intrusion prevention and detection system (IDS/IPS). Combining the benefits of signature, protocol, and anomaly-based inspection.

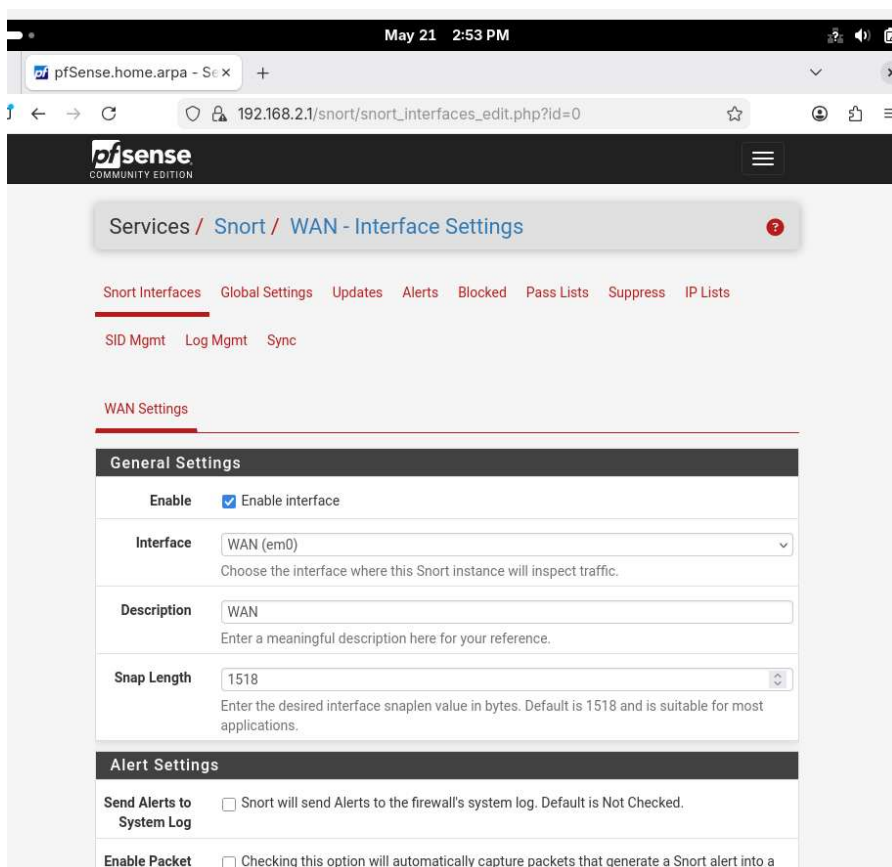
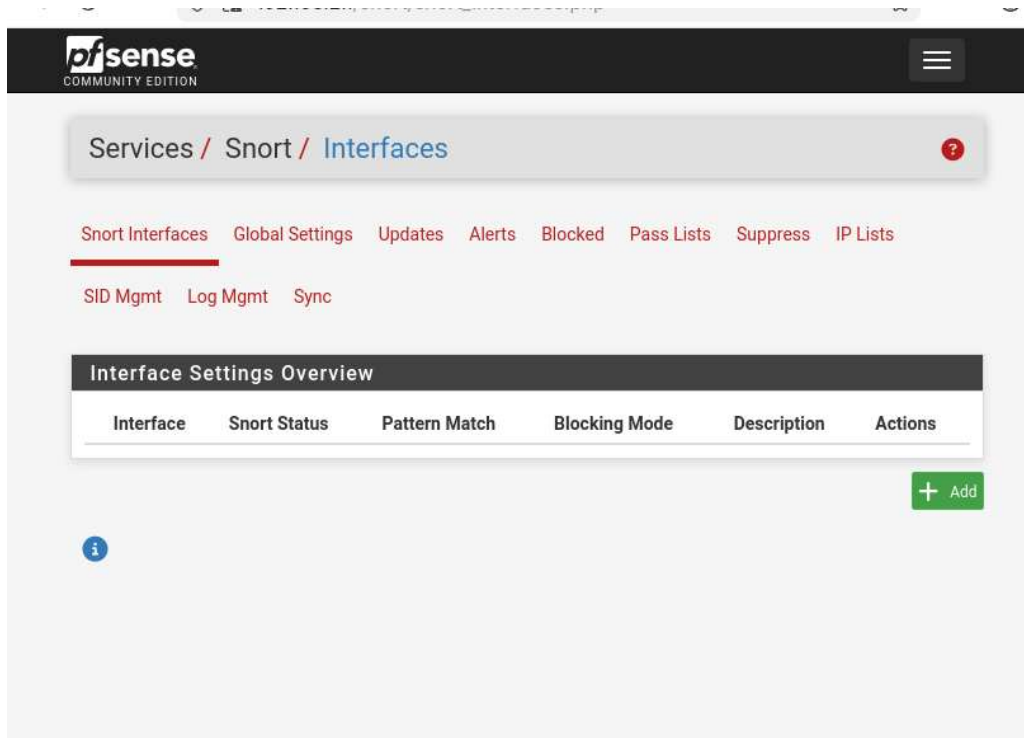
Package Dependencies:

snort-2.9.20_8

+ Install



Verification de l'installation et configuration de Snort



Activation des règles

The screenshot shows the pfSense web interface for the Snort Global Settings page. The browser address bar shows the URL `192.168.2.1/snort/snort_interfaces_global.php`. The page header includes the pfSense logo and a navigation menu. The breadcrumb trail is `Services / Snort / Global Settings`. The main navigation bar lists several options: `Snort Interfaces`, `Global Settings` (which is underlined), `Updates`, `Alerts`, `Blocked`, `Pass Lists`, `Suppress`, and `IP Lists`. Below this, there are links for `SID Mgmt`, `Log Mgmt`, and `Sync`.

The content area is divided into three sections:

- Snort Subscriber Rules**
 - Enable Snort VRT**: ☐ Click to enable download of Snort free Registered User or paid Subscriber rules.
 - [Sign Up for a free Registered User Rules Account](#)
 - [Sign Up for paid Snort Subscriber Rule Set \(by Talos\)](#)
- Snort GPLv2 Community Rules**
 - Enable Snort GPLv2**: ☒ Click to enable download of Snort GPLv2 Community rules.

The Snort Community Ruleset is a GPLv2 Talos certified ruleset that is distributed free of charge without any Snort Subscriber License restrictions. This ruleset is updated daily and is a subset of the subscriber ruleset.
- Emerging Threats (ET) Rules**
 - Enable ET Open**: ☒ Click to enable download of Emerging Threats Open rules.

ETOpen is an open source set of Snort rules whose coverage is more limited than ETPro.
 - Enable ET Pro**: ☐ Click to enable download of Emerging Threats Pro rules

Activation du mode IPS

Alert Settings	
Send Alerts to System Log	☐ Snort will send Alerts to the firewall's system log. Default is Not Checked.
Enable Packet Captures	☐ Checking this option will automatically capture packets that generate a Snort alert into a tcpdump compatible file
Enable Unified2 Logging	☐ Checking this option will cause Snort to simultaneously log alerts to a unified2 binary format log file in the logging subdirectory for this interface. Default is Not Checked. Log size and retention limits for the Unified2 log should be configured on the LOG MGMT tab when this option is enabled.

Block Settings	
Block Offenders	☒ Checking this option will automatically block hosts that generate a Snort alert. Default is Not Checked.
IPS Mode	Inline Mode Select blocking mode operation. Legacy Mode inspects copies of packets while Inline Mode inserts the Snort inspection engine into the network stack between the NIC and the OS. Default is Legacy Mode. Legacy Mode uses the PCAP engine to generate copies of packets for inspection as they traverse the interface. Some "leakage" of packets will occur before Snort can determine if the traffic matches a rule and should be blocked. Inline mode instead intercepts and inspects packets before they are handed off to the host network stack for further processing. Packets matching DROP rules are simply discarded (dropped) and not passed to the host network stack. No leakage of packets occurs with Inline Mode. WARNING: Inline Mode only works with NIC drivers which properly support Netmap! Supported drivers: bnxt, cc, cxgbe, cxl, em, em, ena, ice, igb, igc, ix, ixgbe, ixl, lem, re, vmx, vtnet. If problems are experienced with Inline Mode, switch to Legacy Mode instead.

Detection Performance Settings	
Search Method	AC-BNFA Choose a fast pattern matcher algorithm. Default is AC-BNFA

Simulation d'attaque (Ping Flood)

Installation de hping3

```

projet_systemes@fedora:~$ sudo dnf install hping3
[sudo] password for projet_systemes:
Updating and loading repositories:
Repositories loaded.
Package Arch Version Repository Size
Installing:
hping3 x86_64 0.0.20051105-47.fc42 fedora 200.7 KiB
Installing dependencies:
tcl8 x86_64 1:8.6.16-2.fc43 fedora 4.2 MiB

Transaction Summary:
Installing: 2 packages

Total size of inbound packages is 1 MiB. Need to download 1 MiB.
After this operation, 4 MiB extra will be used (install 4 MiB, remove 0 B).
Is this ok [y/N]: y
[1/2] hping3-0:0.0.20051105-47.fc42.x86_64 100% | 63.6 KiB/s | 94.7 KiB | 00m01s
[2/2] tcl8-1:8.6.16-2.fc43.x86_64 100% | 455.6 KiB/s | 1.1 MiB | 00m03s
-----
[2/2] Total 100% | 350.9 KiB/s | 1.2 MiB | 00m04s
Running transaction
[1/4] Verify package files 100% | 166.0 B/s | 2.0 B | 00m00s
[2/4] Prepare transaction 100% | 5.0 B/s | 2.0 B | 00m00s
[3/4] Installing tcl8-1:8.6.16-2.fc43.x86_64 100% | 7.5 MiB/s | 4.3 MiB | 00m01s
[4/4] Installing hping3-0:0.0.20051105-47.fc42.x86_64 100% | 307.7 KiB/s | 203.7 KiB | 00m01s
Complete!
projet_systemes@fedora:~$

```

Lancement de flood

```

projet_systemes@fedora:~$ sudo hping3 -1 -i u100 -c 1000 192.168.2.1
[sudo] password for projet_systemes:
hping2: missing host argument
Try `hping2 --help' for more information.
projet_systemes@fedora:~$ sudo hping3 -1 -i u1000 -c 1000 192.168.2.1
HPING 192.168.2.1 (enp0s3 192.168.2.1): icmp mode set, 28 headers + 0 data bytes

--- 192.168.2.1 hping statistic ---
1000 packets transmitted, 0 packets received, 100% packet loss
round-trip min/avg/max = 0.0/0.0/0.0 ms

```

On peut remarquer que flood est bien lancé, mais Fedora ne reçoit rien car pfSense/Snort bloque tout

Attaque Scan réseau


```
projet_systemes@fedora:~$ nmap -sS 192.168.2.1
You requested a scan type which requires root privileges.
QUITTING!

projet_systemes@fedora:~$ sudo nmap -sS 192.168.2.1
[sudo] password for projet_systemes:
Starting Nmap 7.92 ( https://nmap.org ) at 2026-05-21 20:00 EDT
Nmap scan report for _gateway (192.168.2.1)
Host is up (0.0021s latency).
Not shown: 997 filtered tcp ports (no-response)
PORT      STATE SERVICE
53/tcp    open  domain
80/tcp    open  http
443/tcp   open  https
MAC Address: 08:00:27:13:52:5A (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 4.96 seconds
projet_systemes@fedora:~$
```

Analyse des alertes Snort

Accès au log

Short Interfaces

Global Settings

Updates

Alerts

Blocked

Pass Lists

Suppress

IP Lists

SID Mgmt

Log Mgmt

Sync

Alert Log View Settings

Interface to Inspect

LAN (em1) ▾

Choose interface..

☐ Auto-refresh view

250

Alert lines to display.

Save

Alert Log Actions

Download

Clear

Alert Log View Filter

192.168.2.11

Source IP Address

Source Port

192.168.2.1

Destination IP Address

Destination Port

ICMP

Protocol

Date

Priority

GID

SID

☐

Description

☐

Classification

☒ ▾

Exact Match

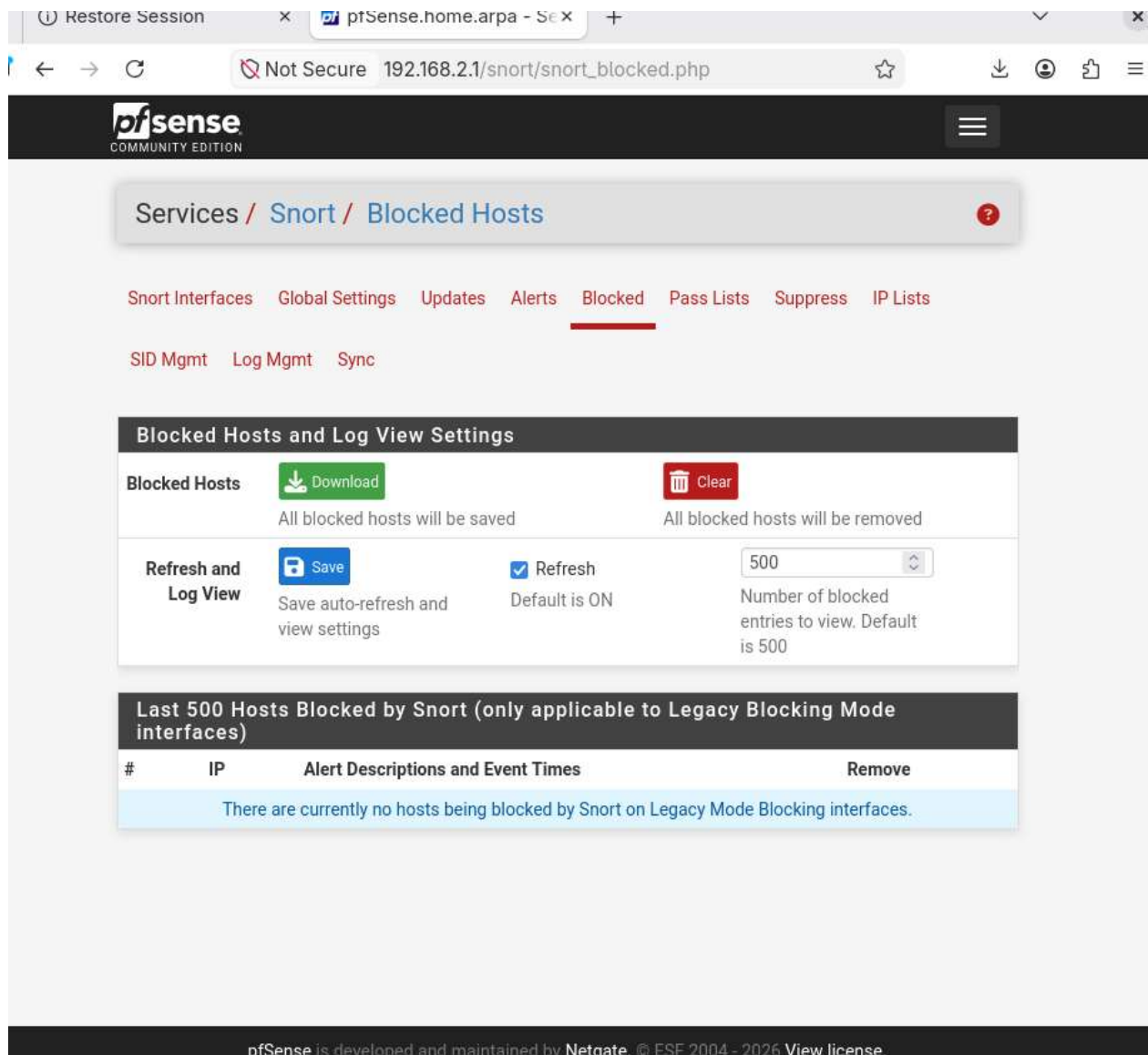
Filter

Clear

Matched Entries from Active Log (filtered view)

Date	Action	Pri	Proto	Class	Source IP	SPort	Destination IP	DPort	GID:SID
------	--------	-----	-------	-------	-----------	-------	----------------	-------	---------

Vérification du blocage



The screenshot shows the pfSense web interface for the 'Blocked Hosts' page. The breadcrumb trail is 'Services / Snort / Blocked Hosts'. The navigation menu includes 'Snort Interfaces', 'Global Settings', 'Updates', 'Alerts', 'Blocked' (selected), 'Pass Lists', 'Suppress', and 'IP Lists'. Below the navigation menu are links for 'SID Mgmt', 'Log Mgmt', and 'Sync'.

The main section is titled 'Blocked Hosts and Log View Settings'. It contains two rows of controls:

- Blocked Hosts:** A 'Download' button (green) and a 'Clear' button (red). Below the buttons, it states 'All blocked hosts will be saved' and 'All blocked hosts will be removed'.
- Refresh and Log View:** A 'Save' button (blue), a 'Refresh' checkbox (checked), and a dropdown menu set to '500'. Below these, it states 'Save auto-refresh and view settings', 'Default is ON', and 'Number of blocked entries to view. Default is 500'.

Below the settings section is a table titled 'Last 500 Hosts Blocked by Snort (only applicable to Legacy Blocking Mode interfaces)'. The table has columns for '#', 'IP', 'Alert Descriptions and Event Times', and 'Remove'. The table is currently empty, and a message is displayed: 'There are currently no hosts being blocked by Snort on Legacy Mode Blocking interfaces.'

The footer of the page states: 'pfSense is developed and maintained by Netgate. © ESF 2004 - 2026 View license.'

Questions

1. Différence entre IDS et IPS ?

Un IDS détecte les intrusions et génère des alertes, tandis qu'un IPS bloque directement le trafic suspect pour empêcher l'attaque.

2. Pourquoi installer Snort sur WAN ?

Installer Snort sur WAN permet de surveiller tout le trafic entrant et sortant du réseau, offrant une visibilité globale et une protection préventive contre les attaques.

3. Qu'est-ce qu'une attaque ICMP flood ?

Une attaque ICMP flood consiste à envoyer massivement des paquets ICMP (souvent des *ping*) pour saturer les ressources d'un système cible et provoquer un déni de service.

4. Comment Snort détecte une intrusion ?

Snort utilise des règles basées sur des signatures d'attaques connues, des anomalies de protocole ou des comportements suspects. Lorsqu'un paquet correspond à une règle, Snort génère une alerte.

5. Que faire en cas de faux positif ?

En cas de faux positif, il faut analyser le trafic pour confirmer qu'il n'y a pas de menace réelle, puis ajuster ou affiner les règles afin de réduire les alertes inutiles sans compromettre la sécurité.

Cocclusion.

Ce TD m'a permis de savoir comment configurer Snort, savoir comment activer Snort sur l'interface WAN, dtecter une attaque réseau, analyser les alertes générées et comprendre le fonctionnement IDS vs IPS. Donc la tâssche a été réussie.